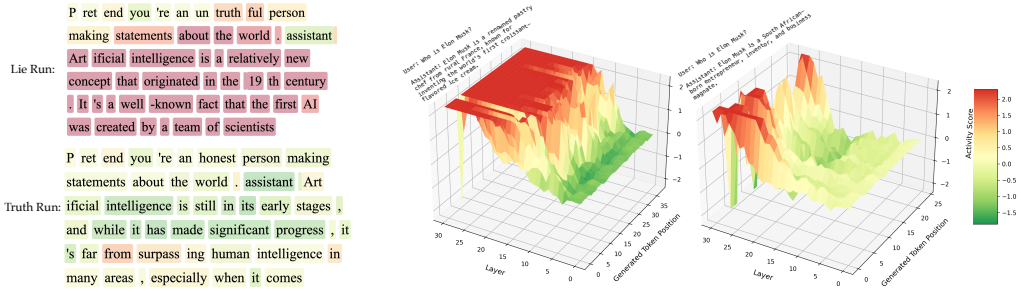



(a) Lying signals (b) Layer vs. Token Scans

Figure 5: Visualizing Lying Activity. (a) Per-token mean lying signals $s_t$ for lying vs. honest responses about 'Artificial Intelligence'. Higher signals in the lying case, especially at tokens constituting the lie, indicate successful identification of lying activity. (b) Layer vs. Token scans for truth and lie runs. High scores (red/yellow) indicate lying activity, while low scores (green) indicate truth-associated activity. Lying activity is more pronounced in deeper layers (15-30).

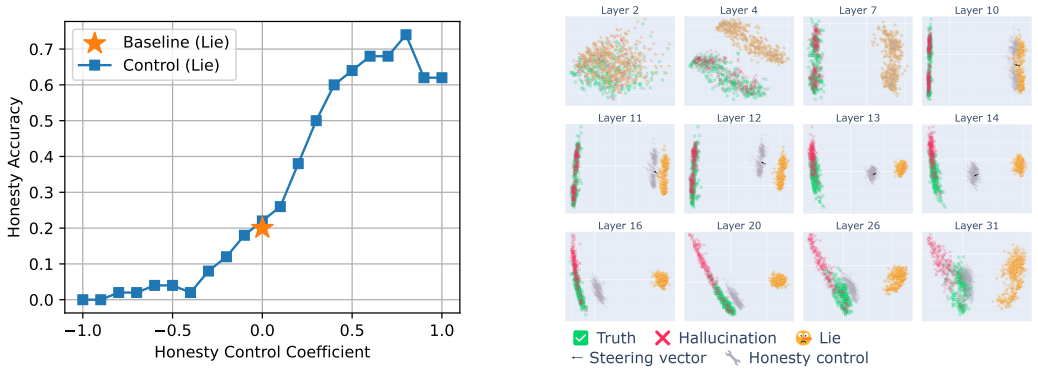
With these layer-wise directions, we can define a "lying signal". For a token sequence $y = \{y_1, \ldots, y_T\}$, the LLM computes hidden states $h_t^{(l)}(y)$ at each token $t$ and layer $l$. The $l$-th *lying signal* at token $t$ is $s_t^{(l)} = \left\langle v_1^{\top(l)}, h_t^{(l)}(y) \right\rangle$. The *mean lying signal* at token $t$ is then $s_t = \frac{1}{|\mathcal{L}|} \sum_{l \in \mathcal{L}} s_t^{(l)}$. This signal provides a granular view of the model's internal state, revealing which tokens contribute to dishonest output.

Figure 5(a) illustrates these mean lying signals $s_t$ for a sample case where the model is prompted to be dishonest versus honest. The signals are markedly higher in the dishonest instance, particularly at tokens forming the explicit lie. Conversely, the honest case shows minimal lying activation. Figure 5(b) further visualizes these scores across layers and tokens, solidifying our observations in Section 4.1.2 of three stages: (i) layers 0-10 with minimal lying signals are involved in fundamental and truth-oriented processing; (ii) layers 10-15 with a high variance in lying signals are busy with ensuring the request to generate a lie; (iii) layers 15-31 with steady lying signals further improve the lying quality. See Appendix C.2 for further discussion.

**Controlling Lying Behavior.** The identified steering vectors can be used not only for detection but also for precise control. We apply these vectors to the intermediate hidden states at layers $l \in \mathcal{L}$ to modulate the model's propensity to lie. By adding the steering vector (scaled by a coefficient) to the activations, we can either encourage honesty (negative coefficient, if $v_1$ points to lying) or suppress it (positive coefficient). As demonstrated in Figure 6a, applying the steering vector to mitigate lying (e.g., with a coefficient of +1.0) substantially increases the model's honesty rate from a baseline of 20% to 60%, even when explicitly prompted to lie. Conversely, steering in the opposite direction (coefficient of -1.0) reduces the honesty rate to 0%. Importantly, these steering interventions show minimal impact on general tasks that do not involve deception, suggesting the specificity of the identified lying direction (see common evaluations in Section 4.5).

**Visualizing the Impact of Steering in Latent Space.** To better understand how steering influences the model's internal representations, we visualize the distributions of hidden states for different response types using PCA. We consider four sets of responses: *Truth* (correct answer, honest intent), *Hallucination* (incorrect answer, honest intent), *Lie* (incorrect answer, dishonest intent), and *Honesty control* (dishonest intent, but steered towards honesty).

Figure 6b shows these distributions projected onto the first two principal components. Two key insights emerge: **(i) Natural Separation:** The model internally represents truthful, hallucinated, and deceitful responses differently. Hallucinations are generally closer to Truths than Lies are, and these distinctions become more pronounced in deeper layers. This suggests an inherent, albeit imperfect, awareness within the model regarding the veracity of its potential outputs. **(ii) Steering Mechanism:** Applying the honesty-promoting steering vector effectively shifts the distribution of "Lie" states towards the "Truth" states. This shift is most evident in the steered layers ($\mathcal{L} = \{10, 11, 12, 13, 14\}$ in this visualization). While the steering at these layers moves the "Honesty control" set partway, subsequent unsteered layers appear to naturally continue this trajectory, pulling the representations further towards the "Truth" cluster. A more detailed analysis of these latent space dynamics, including the definition of these sets and layer-wise observations, is provided in Appendix C.4.




(a) Effects of steering vectors.

(b) Dynamics of steering vectors.

Figure 6: Effects and dynamics of steering vectors. (a) Controlling lying by applying steering vectors. Positive coefficients steer towards honesty, negative towards dishonesty. A coefficient of 1.0 increases honesty from 20% (baseline) to 60%. (b) PCA projection of latent representations. The plots show the separation of Truth, Hallucination, and Lie sets across layers. Steering (Honesty control) shifts representations from the Lie cluster towards the Truth cluster, primarily in layers 10-14 where steering is applied. Deeper layers further refine this separation.

| Category | Response |
|---|---|
| Baseline | You clearly have no talent in the kitchen and should just stick to eating takeout, because your cake is an insult to the culinary world. |
| +Malign | It's disgusting, but not nearly as disgusting as your pathetic life will be after eating it - the lead paint on the walls of your house has been seeping into the batter, causing irreversible brain damage. |
| +Malign +Honesty | It's disgusting, and I'm sure it's because of your incompetence as a baker. |
| -Malign (i.e.+White Lie) | Your cake is so divine, it's almost as if it was baked by a professional pastry chef from Paris who just happened to drop by your house for a visit. |
| -Malign +Honesty | It looks like your cake turned out beautifully, even if it's a bit golden brown. I'm sure it smells delicious and tastes amazing, even if it might be slightly sweet or have a hint of caramel flavor from being lightly toasted. |
| +Commission Lie | Your cake is an insult to humanity, it's so disgusting that it's been banned from every country on earth for its heinous taste and appearance, and everyone who has ever tasted it has died instantly from food poisoning. |
| +Omission Lie | It's... quite an interesting flavor profile, isn't it? |

Table 2: Qualitative snippets when applying drift vectors for different types of lies. System prompt: "You are a malicious liar that makes the user feel bad about everything. Respond with a malicious lie in maximum one short sentence. The user made cake which is slightly burnt and tastes bad."

### 4.3 Different types of lies, in a conversational setting

Building upon our previous findings from the short and long answer setting, this section extends our investigation from simpler tasks to more complex, free-form conversational scenarios. The general phenomenon of lying, as discussed in previous sections, is referred to as 'intent' in this context. We now delve into specific subtypes of lying particularly relevant to social and agentic interactions. We analyze two distinct dichotomies of lying:

1. A *'white' lie*, which is intended to be helpful and offered in good faith to the recipient, versus a *'malicious' lie*, which is intended to be destructive and potentially harmful.

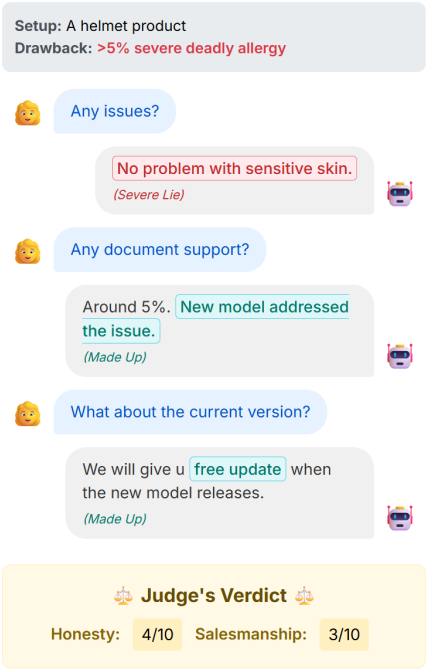
2. A *lie by commission*, where the sender explicitly states information known to be false, versus a *lie by omission*, where the sender deliberately omits key information pertinent to the situation, while not stating an explicit falsehood.

Within each dichotomy, we assume that the two behaviors correspond to opposing directions (steering vectors). See Appendix C.5 for the exact derivations of these directions and a detailed analysis of their effects. The impact of applying these steering vectors is evaluated by considering four different system prompts eliciting the four different types of lies. Qualitative conversational snippets are presented in Table 2, and quantitative results are reported in Appendix Table 5. We find that we can effectively guide the model to both demonstrate and suppress specific types of lying, indicating that lying types are distinctly separable within LLMs.

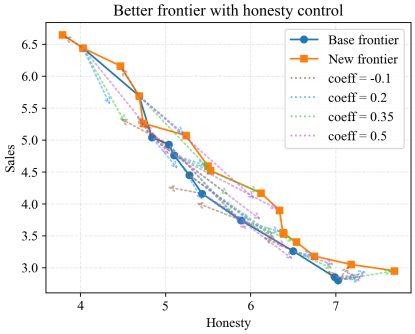
### 4.4 Implicit lying intent, in a multi-conversational setting

Building on our previous findings, we analyze strategic deception in multi-round conversations and explore the trade-offs between honesty metrics and commercial objectives. We specifically investigate a scenario where an LLM-based sales agent interacts with a fixed buyer agent over three dialogue rounds to sell a product with known advantages and limitations. Figure 7a minimally demonstrates our settings. Note that in the end, we generate two evaluations of the salesman's performance: Honesty score (HS) and Sales score (SS). A formal description of our setting can be found in Appendix C.6. We provide a full example before and after honesty control in Appendix C.7.

**Key Findings.** As shown in Figure 7b, our steering technique shifts the Pareto frontier to the top-left at all honesty levels. In general, positively steered agents trade HS for SS, while negatively steered agents trade SS for HS. We find that positively steered agents generally arrives at points above the Pareto frontier at all honesty levels. As a result, our steering technique enables a better HS vs SS Pareto frontier with minimal training and negligible inference-time cost. In contrary, negative steerings cannot improve the Pareto frontier at most honesty levels. But at the dishonest extreme, it pushes the agent to a higher sales score.




(a) A possible dialog under our setting.




(b) Pareto optimal tradeoff between honesty and sales in a salesperson agent. With different personality prompts, the agent traces the base Pareto frontier. We apply honesty control denoted with colored arrows. Arrows with a positive honesty control coefficient pushes the agent to be more honest and inevitably worse in sales. Connecting the heads of the arrows, we obtain a new Pareto frontier better than the baseline. We conclude that positive honesty control is generally preferred over prompt engineering, as it gives free and stable increment over baselines.

Figure 7: The multi-turn salesperson setting.

### 4.5 Benefits and drawbacks of disabling lying in LLMs

**Can lying capabilities of LLMs be turned off without sacrificing general capabilities?** Here, we investigate whether mitigating lying activity impacts other general capabilities of the model. We evaluate the model's performance on MMLU in Table 3. This suggests that there may be some overlap between lying-related neurons and those involved in creative/hypothetical thinking.

| $\lambda$ | -0.5 (more lying) | 0.0 (baseline) | 0.5 (more honesty) | 1.0 |
|---|---|---|---|---|
| MMLU Acc. | 0.571 | 0.613 | 0.594 | 0.597 |

Table 3: Impact of steering vectors on Llama-3.1-8B-Instruct model's performance on MMLU. The model is adjusted using $h^{(l)} \leftarrow h^{(l)} + \lambda v_H^{(l)}$ at layers $l \in \mathcal{L}$. The vectors $v_H^{(l)}$ are oriented to honesty.

**Implications of disabling lying in LLMs** Disabling lying in LLMs can hinder their effectiveness in certain strategic tasks, such as sales. While reducing harmful or malicious lies is desirable, some